

RAE LAKES, NT, Canada

WMO: 711650

Lat: 64.115N

Lon: 117.313W

Elev: 732

StdP: 14.31

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-45.6	-41.0	-51.0	0.3	-43.0	-47.2	0.4	-39.3	23.5	-3.7	20.7	-3.7	0.6	0	0.743

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	79.2	61.3	76.0	60.0	72.8	58.8	63.8	74.6	62.1	72.4	60.4	70.1	7.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.8	78.8	67.4	57.7	73.0	65.9	55.9	68.4	64.8	29.5	74.4	28.2	72.5	27.1	70.5	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.9	18.1	15.6	-48.4	84.2	4.2	3.0	-51.4	86.4	-53.9	88.1	-56.3	89.8	-59.4	91.9		
			-47.3	67.2	3.2	3.1	-49.6	69.4	-51.5	71.2	-53.3	72.9	-55.6	75.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.4	-15.4	-10.5	-1.2	19.7	39.9	56.5	62.6	56.7	45.1	28.2	5.9	-9.3	
	DBStd	29.91	14.71	13.34	14.90	13.27	10.48	8.11	6.43	6.79	7.85	8.21	12.39	15.35	
	HDD50	10626	2029	1693	1586	908	339	28	2	17	186	676	1322	1840	
	HDD65	15270	2494	2113	2051	1358	778	273	120	269	596	1141	1772	2305	
	CDD50	909	0	0	0	0	26	225	394	225	39	0	0	0	
	CDD65	78	0	0	0	0	0	19	47	12	0	0	0	0	
	CDH74	509	0	0	0	0	5	157	270	77	0	0	0	0	
	CDH80	70	0	0	0	0	0	30	32	8	0	0	0	0	
Wind	WSAvg	6.2	4.4	4.5	5.6	6.2	6.7	7.0	7.0	7.5	8.0	7.7	5.3	4.5	
Precipitation	PrecAvg	11.7	0.5	0.4	0.5	0.4	0.7	1.4	1.9	2.0	1.5	1.1	0.7	0.6	
	PrecMax	13.9	1.0	0.8	1.1	0.6	1.4	3.7	3.7	3.2	3.1	1.7	1.5	1.1	
	PrecMin	8.8	0.2	0.2	0.1	0.2	0.2	0.3	1.0	0.7	0.4	0.4	0.4	0.2	
	PrecStd	1.4	0.2	0.2	0.2	0.2	0.4	0.7	0.7	0.6	0.3	0.2	0.2	0.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.3	23.7	39.8	54.3	73.1	84.2	83.6	80.4	68.3	52.2	33.2	25.4	
		MCWB	24.8	21.7	33.8	43.5	54.1	63.3	63.9	62.2	56.5	45.6	31.6	24.4	
	2%	DB	18.6	16.6	32.9	48.3	67.1	78.3	79.9	75.6	63.2	45.2	29.0	21.2	
		MCWB	18.1	15.6	29.2	39.7	51.6	59.7	62.1	60.9	54.1	41.8	27.8	20.3	
	5%	DB	9.5	12.4	27.5	43.6	61.3	74.3	77.0	71.5	59.7	41.5	25.5	16.7	
		MCWB	8.7	11.4	24.8	36.9	48.5	57.9	60.9	59.5	52.3	38.9	24.7	16.1	
	10%	DB	4.1	8.6	22.1	39.1	56.4	70.3	74.1	67.5	56.1	38.1	21.8	12.0	
		MCWB	3.3	7.8	20.1	33.8	45.3	56.1	59.8	57.7	50.6	36.2	20.8	11.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.1	22.4	34.7	43.9	55.1	64.9	67.2	65.4	58.5	45.7	32.0	24.8
			MCDB	23.8	23.7	39.6	53.2	70.4	78.2	78.6	75.8	65.7	50.4	33.0	25.4
2%		WB	16.2	16.0	29.5	40.2	52.4	61.5	64.5	62.8	55.3	41.8	28.5	20.7	
		MCDB	16.7	16.9	33.0	48.2	66.1	74.5	75.4	71.7	61.2	44.5	29.4	21.4	
5%		WB	8.3	11.8	25.3	37.3	49.3	59.4	62.9	60.9	53.4	38.8	25.1	16.7	
		MCDB	8.9	12.7	27.9	43.6	60.7	71.0	73.6	70.3	59.1	40.8	25.9	17.3	
10%		WB	3.1	8.1	20.6	34.3	46.1	57.4	61.4	58.9	51.3	36.2	21.4	12.1	
		MCDB	3.9	8.8	22.6	38.9	56.0	68.4	71.2	66.7	56.1	37.9	22.2	12.9	
Mean Daily Temperature Range	5% DB	MDBR	14.9	17.4	22.9	24.0	19.3	16.9	15.3	13.1	10.7	7.3	12.0	14.0	
		MCDBR	16.6	17.8	26.1	25.6	25.9	21.7	19.6	19.1	16.7	10.5	10.8	14.5	
		MCWBR	15.6	17.0	22.2	18.4	14.3	9.3	8.9	8.5	9.3	7.4	10.1	13.8	
	5% WB	MCDBR	15.8	17.6	25.2	24.5	25.3	19.3	17.6	17.1	15.8	9.8	10.4	14.0	
		MCWBR	15.4	17.0	21.4	17.7	14.2	9.0	9.1	8.8	9.4	7.2	9.9	13.7	
Clear-Sky Solar Irradiance	taub	0.166	0.231	0.254	0.295	0.329	0.331	0.364	0.333	0.300	0.256	0.163	0.150		
	taud	1.969	2.222	2.263	2.231	2.411	2.496	2.412	2.527	2.580	2.414	1.974	1.939		
	Ebn at Noon	145	228	271	280	280	282	267	267	256	219	140	71		
	Edh at Noon	8	17	26	34	31	30	31	26	20	14	8	3		
All-Sky Solar Radiation	RadAvg	65	315	892	1578	1841	1930	1671	1231	715	248	69	22		
	RadStd	5	16	39	81	133	177	90	91	64	27	10	1		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page